18A 1 BENZENE: A MOLECULE WITH TWO MODELS

SPECIFICATION REFERENCE
18.1 18.2 18.3

LEARNING OBJECTIVES

- Be able to use thermochemical, X-ray diffraction and infrared data as evidence for the structure and stability of the benzene ring.
- Understand that the delocalised model for the structure of benzene involves overlap of p-orbitals to form π–bonds.
- Understand why benzene is resistant to bromination, compared to alkenes, in terms of delocalisation of π-bonds in benzene compared to the localised electron density of the π-bond in alkenes.

WHAT ARE AROMATIC COMPOUNDS?

The sections in **Topic 18** deal with compounds containing a benzene ring. You have already seen benzene rings in previous sections. However, no detail was given about them. For example, in the section of **Topic 15** on carbonyl compounds, there was a reaction involving 2,4-dinitrophenylhydrazine, which contains a benzene ring. In the section of **Topic 15** on carboxylic acids, one of the monomers used to make polyesters was terephthalic acid, which also contains a benzene ring.



▲ **fig A** Many brightly coloured dyes are made from aromatic compounds.

ORIGIN OF THE TERM

The word '**aromatic**' in everyday language refers to smells, usually pleasant. Some herbs used in cooking are often described as aromatic.

ALIPHATIC AND AROMATIC

With a few exceptions, almost all the organic compounds you have seen so far in this course could have been described as 'aliphatic', although this is a term we did not use to describe them because there was no point. You can regard aliphatic compounds as all those that are not aromatic!

BENZENE

The compound benzene is at the heart of every aromatic compound. So, the first thing we must do is understand what is special about benzene.

Benzene can be described as an arene, as can many of its derivatives. You already know that *-ene* indicates the presence of a C=C double bond in a molecule (as in ethene). Therefore, you would expect that an arene also contains C=C double bonds. The answer is – yes and no. This obviously needs careful explanation!

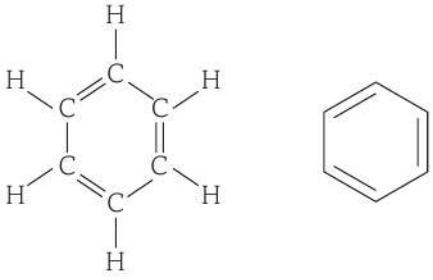
PHYSICAL PROPERTIES OF BENZENE

Benzene is a colourless liquid with a boiling temperature of 80 °C, and it is insoluble in water. It is present in crude oil and the fuels obtained from it, so the petrol tank of a car contains some benzene. The one place you will not find it is in a school or college laboratory, because of its toxic nature; in particular, it is a carcinogen (i.e. it can cause cancer). It was first isolated by the English chemist Michael Faraday in 1825. Within a few years, the compound became known as benzene and its molecular formula was established.

Its molecular formula is C_6H_6, which suggests that it is highly unsaturated, because the alkane with six carbon atoms has 14 hydrogen atoms. Finding a structural formula to fit the molecular formula, and which was supported by studies of its chemical reactions, was a challenge for chemists.

THE KEKULÉ STRUCTURE

One of the better-known stories in the history of chemistry is how the German chemist Friedrich August Kekulé came to suggest the structure of benzene that still bears his name. The story goes like this ... Kekulé was sleeping in an armchair in front of a fire. In the flames he imagined some snake-like molecules dancing, one of which held its own tail and whirled around in the flames. Inspired by this dream about snakes, in 1865 he proposed a cyclic or ring structure for benzene. **Fig B** shows the displayed and skeletal forms of this structure.



Displayed Skeletal

▲ **fig B** Kekulé structure of benzene.

Perhaps Kekulé was also influenced by Josef Loschmidt, who had suggested a ring structure for benzene a few years earlier. Loschmidt's role in developing the structure of benzene is not well remembered. However, you may remember his name from **Book 1** as the source of the symbol *L* for the Avogadro constant.

PROBLEMS WITH THE KEKULÉ STRUCTURE

We still use the original skeletal structure to represent benzene today. However, since it was first used, there has been mounting experimental evidence that does not fit with the structure. Several problems arose in the 19th century, although more became apparent in the 20th century.

PROBLEM 1

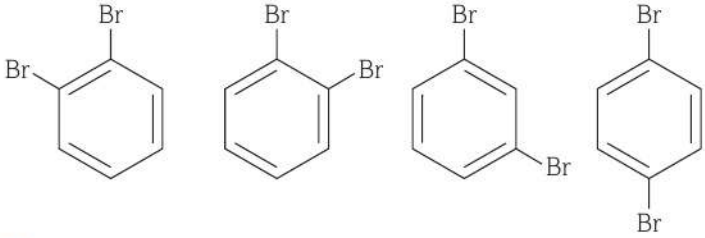
If benzene contains three C=C double bonds, it should readily decolourise bromine water in an addition reaction. However, it does not decolourise bromine water. When bromine does react with benzene, a substitution reaction occurs (see **Section 18A.2**). This evidence suggests that there are no C=C double bonds. **Fig C** shows the results of shaking bromine water with a liquid alkane, benzene and a liquid alkene in separate test tubes.



▲ **fig C** The yellow colour is caused by unreacted bromine.

PROBLEM 2

As more compounds containing the benzene ring were discovered, including those such as dibromobenzene ($C_4H_4Br_2$), another problem arose. If the Kekulé structure is correct, there should be four isomers with this molecular formula:



▲ **fig D** These are the four isomers of $C_6H_4Br_2$, assuming that the Kekulé structure is correct.

However, only three were known to exist. So, the two isomers with bromine on adjacent carbon atoms are identical. It is not the case that adjacent atoms in one isomer are joined by C—C and in the other by C=C. This suggests that the bonds between the carbon atoms in the benzene ring are the same, not different.

PROBLEM 3

When data from X-ray diffraction about the lengths of covalent bonds in molecules became available, there was another problem. **Table A** shows the bond lengths in benzene and cyclohexene (a cyclic alkene with five C—C single bonds and one C=C double bond) in picometres.

BOND	BOND LENGTH/pm
C—C in cyclohexene	154
C=C in cyclohexene	134
C—C in benzene	139

table A

This evidence suggests that the carbon–carbon bonds in benzene are all the same, and perhaps also intermediate in character between C—C and C=C bonds.

PROBLEM 4

When thermochemical data about enthalpy changes of hydrogenation became available, there was yet another problem. **Table B** shows these enthalpy changes for three relevant compounds.

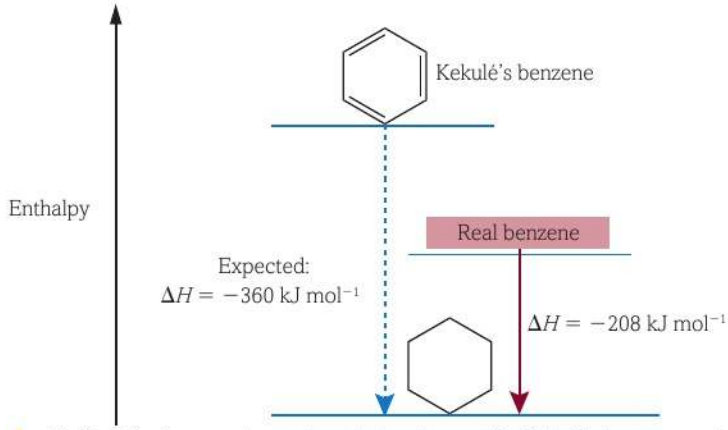
COMPOUND	ENTHALPY CHANGE OF HYDROGENATION/kJ mol^{-1}
cyclohexene	−120
cyclohexa-1,4-diene	−239
cyclohexa-1,3,5-triene (theoretical)	−360
benzene (actual)	−208

table B

All of these hydrogenation reactions form the cyclic alkane cyclohexane. This evidence suggests the following.

- The first two values indicate that the enthalpy change for adding 1 mol of H_2 to 1 mol of C=C bonds is around −120 kJ mol^{-1} (the value for cyclohexa-1,4-diene is double that of cyclohexene because there are twice as many C=C bonds).
- The Kekulé structure could be named cyclohexa-1,3,5-triene, for which the value should be treble that of cyclohexene.
- The actual benzene has a value very much lower (152 kJ mol^{-1} lower) than a theoretical structure with three C=C double bonds would have.

The values for cyclohexa-1,3,5-triene (theoretical) and benzene (actual) can be represented on an enthalpy level diagram, as shown in **fig E**.



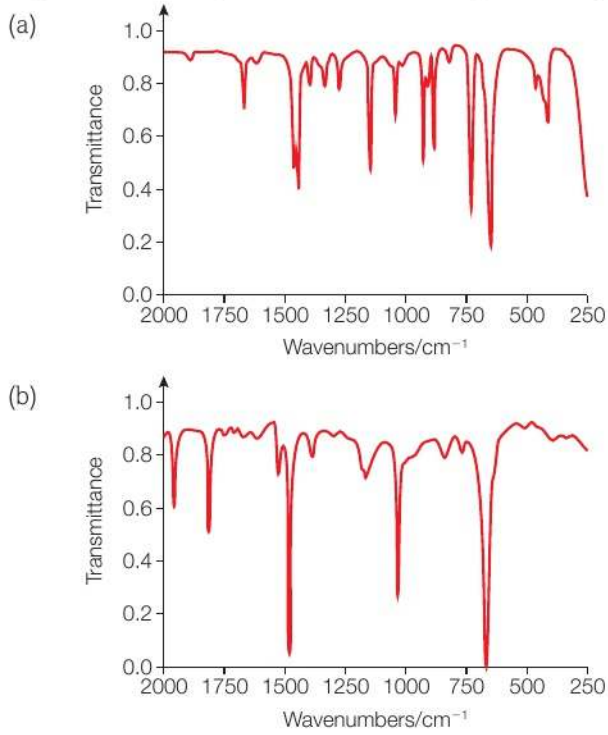
▲ **fig E** This diagram shows the enthalpy changes for Kekulé's benzene and real benzene.

EXAM HINT

When writing an equation for the complete hydrogenation of benzene, make sure you include $3H_2$ to balance the equation (see **Section 18A.2**).

PROBLEM 5

Fig F shows a comparison of the infrared spectra of cyclohexene and benzene.



▲ **fig F** Comparison of the infrared spectra of (a) cyclohexene and (b) benzene.

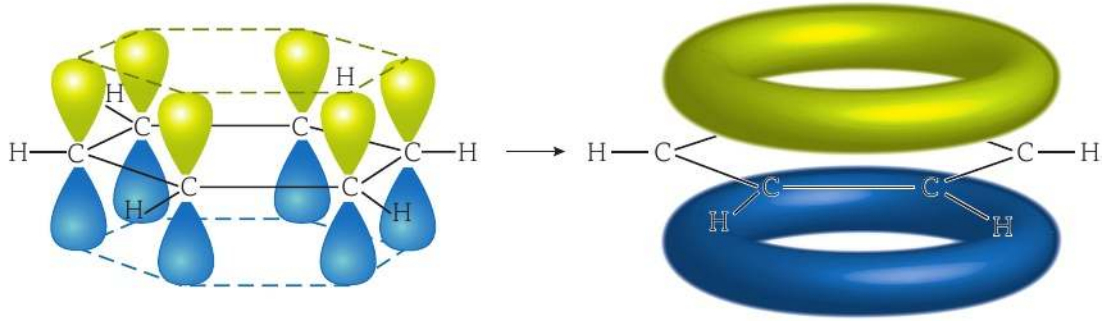
Cyclohexene has an absorption at around 1650 cm^{-1}, which is typical of an isolated alkene C=C stretch (normally between 1680 and 1645 cm^{-1}).

Benzene has a strong absorption at around 1500 cm^{-1}, and also absorptions at 1580 cm^{-1} and 1450 cm^{-1}, which are typical of an aromatic C=C stretch.

A NEW MODEL FOR BENZENE

The weaknesses of the Kekulé structure and growing evidence from other areas of chemistry led to a new model of benzene. In **Book 1**, we explained the C=C double bond in ethene in terms of single sigma C—C bonds and pi bonds formed by the sideways overlap of orbitals.

Now consider the situation in benzene after all the sigma (σ) bonds have formed. Each carbon atom has one electron in a p-orbital. You can imagine the formation of three individual π-bonds, but instead imagine the formation of one large π-bond made up of all six electrons. Three are in a doughnut shape above the atoms and three are in another doughnut shape below the atoms. Together, these six electrons form a delocalised π-bond, as shown in **Fig G**.



▲ **fig G** The delocalised structure of benzene forms when the p-orbitals overlap sideways forming a π-electron cloud above and below the plane of the carbon atoms. Each carbon atom contributes one electron to the cloud, giving a delocalised system of six electrons in total.

PROBLEMS SOLVED?

Let us see how this new model overcomes the problems associated with the Kekulé structure.

Problem 1 There are no individual C=C bonds, so there is no addition reaction with bromine.

Problem 2 There are three, not four, isomers of $C_6H_4Br_2$ because when the bromine atoms are on adjacent carbon atoms there is no difference in the arrangement of electrons between these atoms.

Problem 3 The carbon–carbon bonds are all the same length because they are identical and are not individual C—C and C=C bonds.

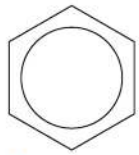
Problem 4 When charge is spread around in a species, there is increased stability, which explains the 152 kJ mol^{-1} greater stability of benzene compared with cyclohexa-1,3,5-triene.

Problem 5 The infrared spectrum of benzene has a very strong absorption at around 1500 cm^{-1}, which is typical of C=C in an aromatic compound. Absorptions for C=C in non-aromatic compounds are between 1680 and 1645 cm^{-1}.

The fact that benzene undergoes a substitution reaction with bromine to form C_6H_5Br, rather than addition to form $C_6H_4Br_2$, can be explained because substitution preserves the stability of the delocalised electrons in the pi bond. The addition reaction would instead produce a compound with two C=C double bonds, and that would lack the extra stability of the product of the substitution reaction.

A NEW MEANING OF 'AROMATIC'

Now we can use the old word '**aromatic**' in a new way – it refers to a hydrocarbon ring containing delocalised electrons. The traditional skeletal formula for benzene can now be replaced by this:



▲ **fig H** The circle represents the delocalised pi bonds with six electrons in total.

LEARNING TIP

When explaining the bonding in a benzene molecule, be careful to avoid confusion between electrons in p-orbitals and delocalised electrons in π-bonds.

CHECKPOINT

1. Using the Kekulé structure for benzene, draw the isomers of tribromobenzene, $C_6H_3Br_3$.
2. How many electrons in a benzene molecule are involved in:
   (a) σ-bonding
   (b) π-bonding?

SUBJECT VOCABULARY

aromatic the original meaning was a description of the smell of certain organic compounds. The new meaning is a description of the bonding in a compound – delocalised electrons forming pi (π) bonding in a hydrocarbon ring

18A 2 SOME REACTIONS OF BENZENE

SPECIFICATION REFERENCE
18.4(i) 18.4(ii) 18.4(iii) 18.4(iv) 18.4(v)

LEARNING OBJECTIVES

- Know the following reactions of benzene, limited to:
  - (i) oxygen in air (combustion to form a smoky flame)
  - (ii) bromine, in the presence of a catalyst
  - (iii) a mixture of concentrated nitric and sulfuric acids
  - (iv) fuming sulfuric acid
  - (v) halogenoalkanes and acyl chlorides with aluminium chloride as catalyst (Friedel–Crafts reaction).

REACTIONS AS A HYDROCARBON

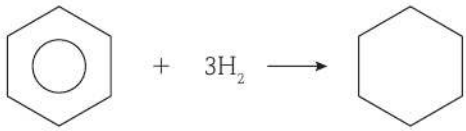
In **Section 18A.1**, hydrogenation was mentioned. This reaction occurs in the same way with benzene as with alkenes. The same is true for combustion.

HYDROGENATION

This is done by mixing benzene with hydrogen and heating under pressure with a nickel catalyst. The equation for the reaction can be shown using molecular or skeletal formulae:

$$C_6H_6 + 3H_2 \rightarrow C_6H_{12}$$

or:



Note the plain hexagon (without a circle) used for the skeletal formula for cyclohexane.

COMBUSTION

As with all hydrocarbons, benzene burns in air, although with a smoky flame. This is typical of many compounds with a high carbon-to-hydrogen ratio. The equation for complete combustion is:

$$C_6H_6 + 7\tfrac{1}{2}O_2 \rightarrow 6CO_2 + 3H_2O$$

REACTIONS AS AN ARENE

The most important reactions of benzene are substitution reactions, in which the products are also aromatic. We will consider the mechanisms of these reactions in the next section (**Section 18A.3**).

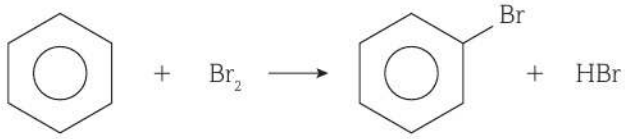
LEARNING TIP

Bromination refers to putting bromine into a molecule. This could be by an addition reaction, but in this section it refers only to a substitution reaction.

BROMINATION

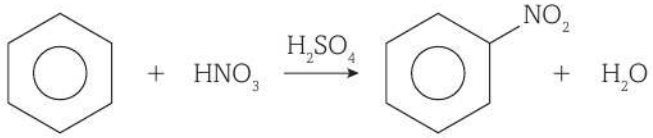
The term 'bromination' has more than one possible meaning. It could refer to the addition reaction between benzene and bromine. This reaction does occur in the presence of ultraviolet radiation, but it is of little importance. In this book, the term refers to a substitution reaction.

Benzene and bromine are heated under reflux, in the presence of a catalyst called a **halogen carrier**. Halogen carriers are usually metal–halogen compounds such as aluminium chloride, aluminium bromide or iron(III) bromide. For bromination, it is enough to add iron filings because they react with some of the bromine to form iron(III) bromide. The products are bromobenzene and hydrogen bromide. The equation for the reaction is:



NITRATION

This refers to the replacement of a hydrogen atom by a nitro group. The nitro group has the formula NO_2 and should not be confused with the gas nitrogen dioxide. The reaction is carried out by warming benzene with a mixture of concentrated nitric and sulfuric acids. The nitric acid can be considered as the source of the NO_2 group and the sulfuric acid as a catalyst. The products are nitrobenzene and water. The equation for the reaction is:



The reaction is carried out between 50 and 60 °C. At lower temperatures the reaction is too slow. At higher temperatures it is possible to form dinitrobenzene, and even trinitrobenzene.

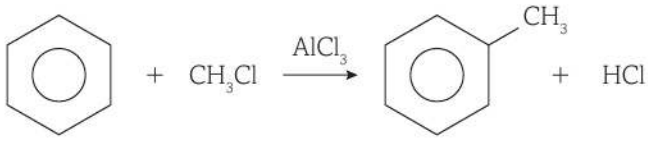
FRIEDEL–CRAFTS REACTIONS

Reactions with this name were developed by Charles Friedel (a French chemist) and James Crafts (an American chemist). There are several types of reaction bearing their names, but we will consider only two of them: alkylation and acylation. They both have these features in common:

- using a reagent represented by XY, one of the hydrogen atoms in benzene is substituted by Y, and the other product is HX
- a catalyst is needed – often aluminium chloride, although other halogen carriers such as iron(III) chloride and iron(III) bromide also work
- anhydrous conditions are needed because water would react with the catalyst and sometimes also with the organic product.

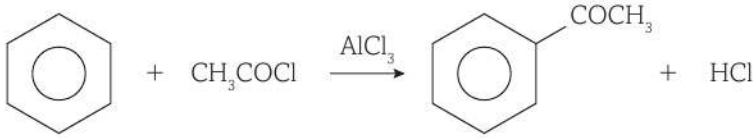
FRIEDEL–CRAFTS ALKYLATION REACTIONS

Alkylation means the substitution of one of the hydrogen atoms of benzene by an alkyl group. The reagent is a halogenoalkane, and the products are an alkylbenzene (methylbenzene) and hydrogen chloride. The equation for the reaction with chloromethane is:



FRIEDEL–CRAFTS ACYLATION REACTIONS

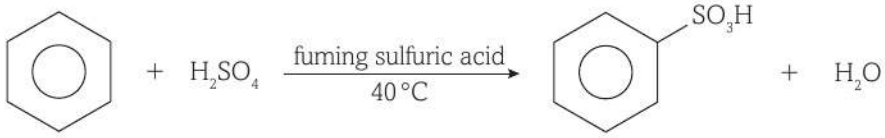
Acylation means the substitution of one of the hydrogen atoms in benzene by an acyl group. The reagent is an acyl chloride, and the products are a ketone (phenylethanone) and hydrogen chloride. The equation for the reaction with ethanoyl chloride is:



phenylethanone

SULFONATION

Sulfonation refers to the replacement of a hydrogen atom by a $-SO_3H$ group. The reaction is carried out by warming benzene with fuming sulfuric acid at 40 °C for 20 to 30 minutes.



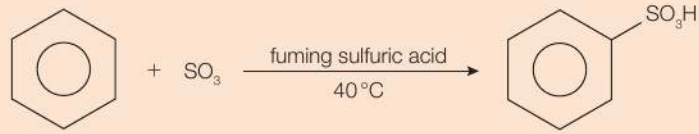
benzenesulfonic acid

The product is benzenesulfonic acid.

Fuming sulfuric acid is made by dissolving sulfur(VI) oxide (sulfur trioxide, SO_3) in concentrated sulfuric acid.

LEARNING TIP

The equation for sulfonation may also be written as



Which equation you use depends on the question you are being asked. If the question is about the reaction with sulfuric acid, then you must use the equation with H_2SO_4. If the question has SO_3 as the electrophile, then you could use this equation.

NAMING AROMATIC COMPOUNDS

The rules for naming aromatic compounds are similar to those used for aliphatic compounds. Prefixes, locants, hyphens and commas, and multiplying prefixes are used in the same way. In some names, you will see 'phen' or 'phenyl' used to represent the benzene ring when it is attached to another functional group. For example, the organic product of the last reaction is phenylethanone: 'ethan' refers to the two carbon atoms in CH_3CO, 'one' indicates that the compound is a ketone and 'phenyl' refers to the C_6H_5 group. You might expect 'benzyl' to be used rather than 'phenyl', but it is not. The explanation for this is beyond the aims of this book.

CHECKPOINT

1. Write an equation for the reaction between benzene and 2-bromopropane.
2. What is the name of the organic product formed in the reaction between benzene and butanoyl chloride?



▲ **fig A** Shoe polish is one of the many commercial products containing aromatic compounds.

SUBJECT VOCABULARY

halogen carrier a compound such as AlX_3 or FeX_3, where X is a halogen, that can catalyse the halogenation of arenes. They produce a stronger electrophile, such as Br^+

18A 3 ELECTROPHILIC SUBSTITUTION MECHANISMS

SPECIFICATION REFERENCE 18.5

LEARNING OBJECTIVES

- Understand the mechanism of the electrophilic substitution reactions of benzene in halogenation, nitration and Friedel–Crafts reactions, including the generation of the electrophile.

ELECTROPHILIC SUBSTITUTION

Although you have not seen this mechanism before, you have studied **substitution** reactions in **Topics 4** and **10 (Book 1: IAS)**, and **electrophilic** addition reactions in **Topic 5 (Book 1: IAS)**.

Before we consider the mechanisms of the individual reactions, it is worth considering what they all have in common:

- The benzene ring is electron-rich. Although it is electrically neutral, the delocalised electrons in the pi (π) bond above and below the atoms mean that the molecule attracts **electrophiles** – species attracted to negative charge.
- Using Y^+ to represent the electrophile, as Y^+ approaches the delocalised π-bond, it attracts two of the six electrons, which form a covalent bond with it. This gives an intermediate species with a positive charge (shown by convention in the centre of the hexagon), in which one carbon atom is joined to both Y and H. There are now only four delocalised electrons, and these are represented by two-thirds of a circle.
- This intermediate species lacks the stability of an aromatic compound (because the circle is broken), so in the next step the H leaves as H^+ and the two electrons in the C—H bond join the remaining four to restore the delocalised π-bond (with its relatively high stability).

When drawing these mechanisms, or skeletal formulae, a substituent on the benzene ring can be shown in any of the six positions. It is usually shown at the top of the hexagon or the next position clockwise.

To help you see the similarities and differences in these mechanisms, they are all set out in the same way.

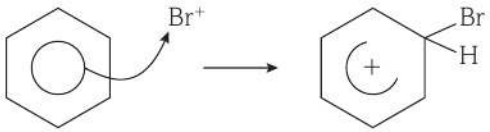
BROMINATION

Step 1: formation of the electrophile

Bromine reacts with the catalyst to form Br^+:

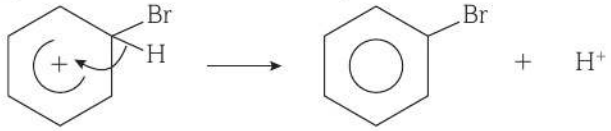
$$AlBr_3 + Br_2 \rightarrow Br^+ + AlBr_4^-$$

Step 2: electrophilic attack



Note that the H atom shown in the product of this step is already present in benzene, although it is not shown. It is shown here because it will be lost in Step 3.

Step 3: formation of the aromatic product



Step 4: regeneration of the catalyst

$$AlBr_4^- + H^+ \rightarrow AlBr_3 + HBr$$

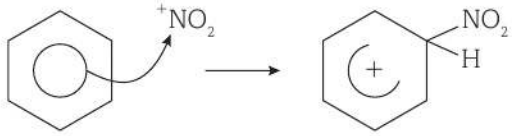
NITRATION

Step 1: formation of the electrophile

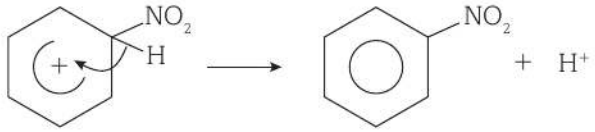
Nitric acid and sulfuric acid react to form NO_2^+ (the nitryl ion or nitronium ion):

$$HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$$

Step 2: electrophilic attack



Step 3: formation of the aromatic product



Step 4: regeneration of the catalyst

$$HSO_4^- + H^+ \rightarrow H_2SO_4$$

EXAM HINT

The formation of the electrophile in this reaction involves nitric acid acting as a *base*. This is because it accepts a proton from the sulfuric acid (see **Topic 14**).

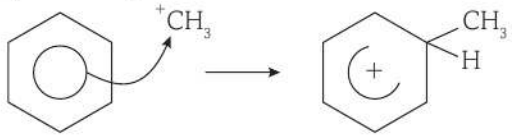
ALKYLATION

Step 1: formation of the electrophile

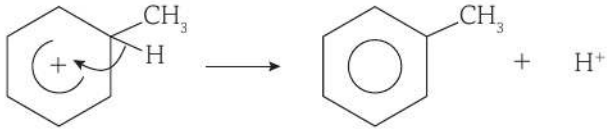
Chloromethane reacts with the catalyst to form CH_3^+:

$$AlCl_3 + CH_3Cl \rightarrow CH_3^+ + AlCl_4^-$$

Step 2: electrophilic attack



Step 3: formation of the aromatic product



Step 4: regeneration of the catalyst

$AlCl_4^- + H^+ \rightarrow AlCl_3 + HCl$

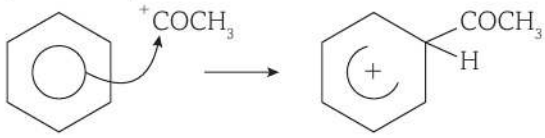
ACYLATION

Step 1: formation of the electrophile

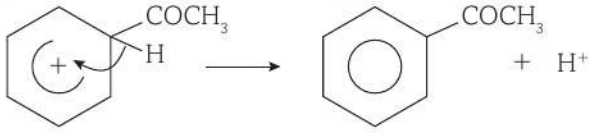
Ethanoyl chloride reacts with the catalyst to form CH_3CO^+:

$AlCl_3 + CH_3COCl \rightarrow CH_3CO^+ + AlCl_4^-$

Step 2: electrophilic attack



Step 3: formation of the aromatic product



Step 4: regeneration of the catalyst

$AlCl_4^- + H^+ \rightarrow AlCl_3 + HCl$

EXAM HINT

If you are asked to show the mechanism for an alkylation or acylation reaction, make sure you show the formation of the electrophile as the first step. Many students miss this step out.

LEARNING TIP

Note the similarities between all these mechanisms. The first step involves formation of the electrophile, and the last step shows what happens to the replaced H^+. The second and third steps show the actual substitution in the benzene ring.

CHECKPOINT

SKILLS DECISION-MAKING

1. Write an equation for Step 1 in the bromination reaction in which the halogen carrier is iron(III) bromide.
2. Write equations for Steps 2 and 3 in the alkylation reaction using 1-bromopropane as a starting material.

SUBJECT VOCABULARY

substitution a reaction in which an atom, or group of atoms, in a molecule is replaced by another atom, or group of atoms

electrophilic a reaction mechanism in which the attacking species is an electrophile

electrophile an atom, molecule or ion that can accept a pair of electrons

18A 4 PHENOL

SPECIFICATION REFERENCE 18.6

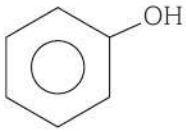
LEARNING OBJECTIVES

- Understand the reaction of phenol with bromine water and the reasons for the relative ease of this reaction compared to benzene.

WHAT IS PHENOL?

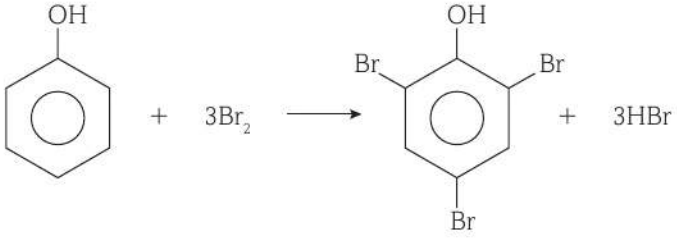
Most aromatic compounds have a benzene ring and an aliphatic part. In previous sections of this topic you have covered some important reactions of the benzene ring. You know many of the reactions of aliphatic compounds from **Topic 15** and from **Book 1**. In this section we will look at one more aromatic compound with a familiar aliphatic group, the hydroxyl (OH) group.

Phenol consists of a hydroxyl group joined to a benzene ring. It has some properties similar to an aliphatic alcohol such as ethanol, and you would expect it to undergo electrophilic substitution reactions because it contains a benzene ring. Its structure is:



BROMINATION OF PHENOL

You will remember from **Section 18A.2** that benzene undergoes bromination, but the reaction needs a catalyst and the reaction mixture has to be heated under reflux. In contrast, the bromination of phenol occurs at room temperature without a catalyst, and works with bromine water. The bromine water is decolourised during the reaction, and the organic product forms as a white precipitate. The equation for the reaction is:

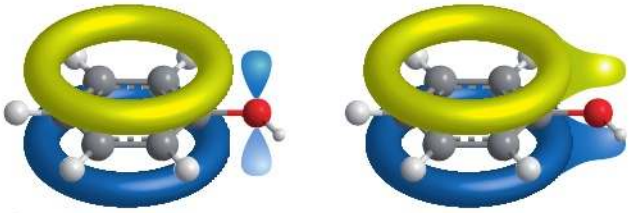


Another obvious difference between benzene and phenol is that the substitution reaction occurs three times, as there are three bromine atoms in the organic product. You can probably see why the name is 2,4,6-tribromophenol. The carbon joined to OH is carbon 1, the other carbons are numbered clockwise, and the three bromine atoms are joined to carbons 2, 4 and 6 in the ring.

It is beyond the aims of this book to try to explain why the reaction involves substitution by three bromine atoms and not any other number. However, we can try to explain why the reaction occurs much more readily with phenol than with benzene.

- The lone pair of electrons in the p_z-orbital on the oxygen atom of the OH group interacts with the delocalised pi (π) electrons in the ring (**fig A**).
- The electron density above and below the ring of atoms is increased, a process sometimes referred to as 'activation' because the molecule is now much more reactive towards electrophiles.
- Bromine molecules, although originally non-polar, are polarised as they approach the benzene ring. Eventually, the Br—Br bond breaks and the Br^+ electrophile attacks the benzene ring.

Fig A shows how the delocalised electrons in the π-bond change because of the presence of the oxygen.



▲ **fig A** The electrons in the p_z-orbital of the oxygen atom become part of the delocalised electron system. This increases the electron density above and below the ring and makes the molecule more attractive to electrophiles such as Br^+.

USES OF PHENOL AND ITS DERIVATIVES

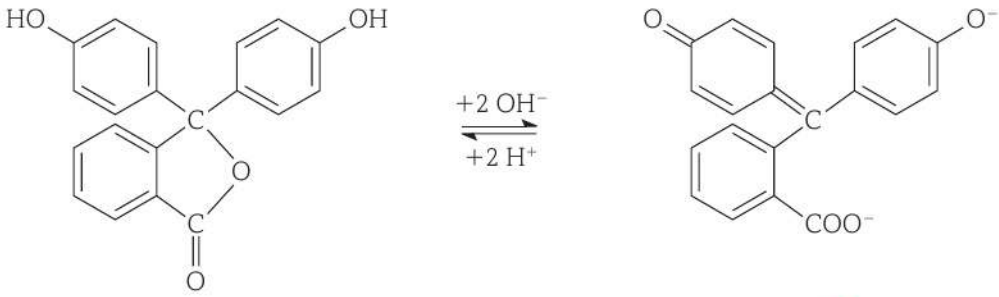
One of the earliest and best known uses of phenol was in antiseptic surgery, especially by the surgeon Joseph Lister. Since then, other antiseptics have been developed from phenol. One of the most familiar of these is the product of the chlorination of phenol – this is 2,4,6-trichlorophenol, better known as the antiseptic TCP. Nowadays, phenol is widely used in many industries, notably in the manufacture of polymers and pharmaceuticals.



▲ **fig B** Antiseptics based on phenol are useful for treating small wounds and grazes.

DID YOU KNOW?

You will remember phenolphthalein as an indicator in acid–base titrations, and will not be surprised to learn that it contains a benzene ring joined to a hydroxyl group! Although you do not need to know its structure, notice that its colour change involves a change in the structure of phenol.



colourless **pink**

▲ **fig C** The phenolphthalein colour change involves a change in the structure of phenol.

The colourless form is a molecule with three benzene rings and two OH groups, while the pink form is an ion with no OH groups and two benzene rings.

LEARNING TIP

Be careful to distinguish between the use of 'benz' and 'phen' in organic names. In general, 'benz' comes after the other groups in the molecule.

CHECKPOINT

1. Suggest why phenol is a solid at room temperature and is slightly soluble in water.
2. Write an equation for the formation of TCP from phenol.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY